



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (1/06/2026 – 7/06/2026)

WEEK 3

PROJECT NAME

ROAD CONDITION DETECTION

PROJECT DESCRIPTION :

**AUTOMATIC DETECTION AND
SEGMENTATION OF ROAD DAMAGES**

GROUP MEMBER :

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GROUP GUIDE :

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1/06/2026	Collected and reviewed additional road damage images to improve underrepresented classes such as rutting, ravelling, line marking, and guard rails.
2/06/2026	Annotated newly collected images and updated the dataset in Roboflow for model retraining
3/06/2026	Annotated newly collected images and updated the dataset in Roboflow for model retraining Annotated newly collected images and updated the dataset in Roboflow for model retraining
4/06/2026	Evaluated model performance using validation metrics and analyzed class-wise detection accuracy for road damage categories.
5/06/2026	Tested the trained model on road videos, generated segmentation outputs, and examined false positives and missed detections.
6/06/2026	Studied association-rule based filtering and designed a two-model pipeline using road segmentation and damage segmentation for improved accuracy.
7/06/2026	Holiday (Sunday)

Next Week Plan	Enhance model accuracy through dataset refinement, association-rule implementation, retraining, and video-based tracking and counting integration.
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REFERENCE:

- RDD2022 Road Damage Dataset (Arya et al., 2024).
- YOLO-based Road Damage Detection Studies (Ma et al., 2025; Wang et al., 2025).
- Ultralytics YOLO Documentation.
- SEA-YOLOv8 Road Surface Damage Detection Framework (Zhao et al., 2025).

SCREEN SHOTS:



